

29/07/2026

- $Z \sim N_n(0, I)$, $A_{n \times n}$ is symm. idempotent.
Then $Z^T A Z \sim \chi^2_{p(A)}$.

* Prediction problem in simple linear regression

$$Y_{\text{new}} = \beta_0 + \beta_1 x_h + \varepsilon_{\text{new}}$$

Point prediction: $\hat{Y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h$

Q) Find a $100(1-\alpha)\%$ prediction interval for Y_{new} .

- Confidence interval for $\beta_0, \beta_1, \beta_0 + \beta_1 x_h$

- We already have developed tests (t-test) for associated hypotheses:

(i) $H_0: \beta_0 = \bar{\beta}_0$ v/s $H_1: \beta_0 \neq \bar{\beta}_0$

(ii) $H_0: \beta_1 = \bar{\beta}_1$ v/s $H_1: \beta_1 \neq \bar{\beta}_1$

(iii) $H_0: \beta_0 + \beta_1 x_h = v_0$ v/s $H_1: \beta_0 + \beta_1 x_h \neq v_0$

- Duality b/w hypothesis testing and confidence intervals (sets).

Ex: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$, $H_0: \theta = \bar{\theta}$ v/s $H_1: \theta \neq \bar{\theta}$

$$A_\alpha(\bar{\theta}) = \{ \underline{x} : |\sqrt{n}(\bar{x} - \bar{\theta})| \leq z_{1-\alpha/2} \} \quad \begin{matrix} \nwarrow \\ 1-\frac{\alpha}{2} \text{ quantile of } N(0,1) \\ = \Phi^{-1}(1-\frac{\alpha}{2}) \end{matrix}$$

$$C_\alpha(\underline{x}) = \{ \theta : |\sqrt{n}(\bar{x} - \theta)| \leq z_{1-\alpha/2} \}$$

Let $\mathcal{P} = \{ P_\theta : \theta \in \Theta \}$ is the family of probability models.
Data $\underline{X} \sim P_\theta$

Test: $H_0: \theta = \bar{\theta}$ v/s $H_1: \theta \neq \bar{\theta} \rightarrow \star$

For each $\bar{\theta} \in \Theta$, let $A_\alpha(\bar{\theta})$ denote a level- α acceptance region for testing $\star$

$$P_{\bar{\theta}}(A_\alpha(\bar{\theta})) \geq 1-\alpha \text{ for all } \bar{\theta} \in \Theta$$

We say that $C_\alpha(\underline{X}) \subset \mathcal{H}$ is a $(1-\alpha)$ -level confidence set for θ if $P_\theta(C_\alpha(\underline{X})) \geq 1-\alpha \quad \forall \theta \in \mathcal{H}$
~~an important~~ $\hookrightarrow P_\theta(\theta \in C_\alpha(\underline{X})) \geq 1-\alpha$ (more explicitly)

By the defⁿs of $A_\alpha(\bar{\theta})$ & $C_\alpha(\underline{X})$, we can say:
 $C_\alpha(\underline{X}) = \{\theta : \underline{X} \in A_\alpha(\theta)\} \quad \forall \theta \in \mathcal{H}$

→ Confidence intervals for $\beta_0, \beta_1, \beta_0 + \beta_1 x_h$.

- For β_0 : $\hat{\beta}_0 \pm t_{n-2}^{1-\alpha/2} \cdot s(\hat{\beta}_0)$
- For β_1 : $\hat{\beta}_1 \pm t_{n-2}^{1-\alpha/2} \cdot s(\hat{\beta}_1)$
- For $\beta_0 + \beta_1 x_h$: $\hat{\beta}_0 + \hat{\beta}_1 x_h \pm t_{n-2}^{1-\alpha/2} \cdot s(\hat{\beta}_0 + \hat{\beta}_1 x_h)$
- For $\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$:

Associated test: $H_0: \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} \bar{\beta}_0 \\ \bar{\beta}_1 \end{pmatrix} \quad \text{v/s} \quad H_1: \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \neq \begin{pmatrix} \bar{\beta}_0 \\ \bar{\beta}_1 \end{pmatrix}$

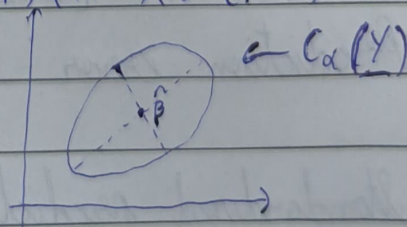
Approach I: (Standardize LS est. of β)

$$W = (\hat{\beta} - \bar{\beta})^T (s^2(\hat{\beta}))^{-1} (\hat{\beta} - \bar{\beta}) \stackrel{H_0}{\sim} 2F_{2, n-2}$$

$$A_\alpha(\bar{\beta}) = \{\underline{y} : W \leq 2F_{2, n-2}^{1-\alpha}\} \rightarrow \text{Acceptance region}$$

Confidence set for β .

$$C_\alpha(\underline{Y}) = \{\beta \in \mathbb{R}^2 : (\beta - \hat{\beta})(X^T X)^{-1}(\beta - \hat{\beta}) \leq 2\hat{\sigma}^2 F_{2, n-2}^{1-\alpha}\}$$



Approach II: (Bonferroni's approach)

$$H_0^{(0)} : \beta_0 = \bar{\beta}_0 \quad \text{v/s} \quad H_1^{(0)} : \beta_0 \neq \bar{\beta}_0$$

$$H_0^{(1)} : \beta_1 = \bar{\beta}_1 \quad \text{v/s} \quad H_1^{(1)} : \beta_1 \neq \bar{\beta}_1$$

$$H_0 = H_0^{(0)} \cap H_0^{(1)}$$

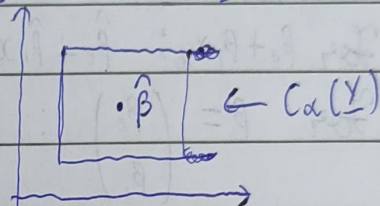
Let $A_{\alpha/2}^{(0)}(\bar{P}_0)$ be $\frac{\alpha}{2}$ -level acceptance region for $H_0^{(0)}$ v/s $H_1^{(0)}$. Same for $A_{\alpha/2}^{(1)}(\bar{P}_1)$.

By Bonferroni procedure, acceptance region for H_0 is $A_{\alpha/2}^{(0)}(\bar{P}_0) \cap A_{\alpha/2}^{(1)}(\bar{P}_1)$

$$= \{ \underline{Y} : |\hat{P}_0 - \bar{P}_0| \leq t_{n-2}^{1-\alpha/4} \cdot s(\hat{P}_0) \text{ \& \> } |\hat{P}_1 - \bar{P}_1| \leq t_{n-2}^{1-\alpha/4} \cdot s(\hat{P}_1) \}$$

Using duality: a $(1-\alpha)$ -level confidence set for $\underline{P}$ is given: $C_\alpha(\underline{Y}) = \{ \underline{\beta} : \bar{Y} \in A_\alpha(\underline{\beta}) \}$

$$= [\hat{P}_0 \pm t_{n-2}^{1-\alpha/4} \cdot s(\hat{P}_0)] \times [\hat{P}_1 \pm t_{n-2}^{1-\alpha/4} \cdot s(\hat{P}_1)]$$



→ Back to problem of prediction:

We want prediction interval for $Y_{\text{new}} = P_0 + P_1 X_h + E_{\text{new}}$
 Confidence interval for $P_0 + P_1 X_h = \theta$

↓

$$C_\alpha(\underline{Y}) = [L(\underline{Y}), U(\underline{Y})] \subset \mathbb{R}$$

$$P_{\underline{\beta}, \sigma^2}(\theta \in C_\alpha(\underline{Y})) = 1 - \alpha \quad \forall \quad \underline{\beta}, \sigma^2$$

$$\tilde{C}_\alpha(\underline{Y}) = [\tilde{L}(\underline{Y}), \tilde{U}(\underline{Y})] \text{ s.t. } P_{\underline{\beta}, \sigma^2}(Y_{\text{new}} \in \tilde{C}_\alpha(\underline{Y})) = 1 - \alpha$$

Prediction error: $Y_{\text{new}} - \hat{Y}_h \sim N(0, \sigma^2 + \sigma^2(\frac{1}{n} + \frac{(X_h - \bar{X})^2}{S_x^2}))$
 $\hookrightarrow \hat{P}_0 + \hat{P}_1 X_h \rightarrow \text{Indp. of SSE}$

Standardized prediction error:

$$\frac{Y_{\text{new}} - \hat{Y}_h}{s(Y_{\text{new}} - \hat{Y}_h)} = \frac{Y_{\text{new}} - \hat{Y}_h}{\hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(X_h - \bar{X})^2}{S_x^2}}} \sim \frac{N(0, 1)}{\sqrt{\frac{X_{n-2}^2}{n-2}}} \sim t_{n-2}$$

$$\Rightarrow P_{e, \sigma^2} \left(\left| \frac{Y_{\text{new}} - \hat{Y}_h}{s(Y_{\text{new}} - \hat{Y}_h)} \right| \leq t_{n-2}^{1-\alpha/2} \right) = 1 - \alpha$$

$$\Rightarrow \tilde{C}_\alpha(Y) = [\hat{Y}_h \pm t_{n-2}^{1-\alpha/2} \cdot s(Y_{\text{new}} - \hat{Y}_h)]$$

* Influence of an individual observation on the LS est.

Let $\{(X_i, Y_i)\}_{i=1}^n$ be the full data and $(\hat{\beta}_0, \hat{\beta}_1)$ denotes the LS est. of $\beta = (\beta_0, \beta_1)$ based on full data.

Let $(\hat{\beta}_0^{(-n)}, \hat{\beta}_1^{(-n)})$ denote the LS est. of β based on $\{(X_i, Y_i)\}_{i=1}^{n-1}$

$$\therefore \hat{\beta}_1 = \hat{\beta}_1^{(-n)} + \frac{X_n - \bar{X}}{\sum_{i=1}^{n-1} (X_i - \bar{X})^2} e_n^{(-n)} \quad \text{where } e_n^{(-n)} = Y_n - \underbrace{(\hat{\beta}_0^{(-n)} + \hat{\beta}_1^{(-n)} X_n)}_{\substack{\text{fitted value for } Y_n \\ \text{based on data} \\ \{(X_i, Y_i)\}_{i=1}^{n-1}}}$$

$$\begin{array}{l|l} \bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X} & \bar{Y} = \frac{(n-1)}{n} \bar{Y}^{(-n)} + \frac{1}{n} Y_n \\ \bar{Y}^{(-n)} = \hat{\beta}_0^{(-n)} + \hat{\beta}_1^{(-n)} \bar{X}^{(-n)} & \bar{X} = \left(\frac{n-1}{n}\right) \bar{X}^{(-n)} + \frac{1}{n} X_n \end{array}$$

$H_0: \beta = \bar{\beta}$ v/s $H_1: \beta \neq \bar{\beta}$

Try to relate $(\hat{\beta} - \bar{\beta})^T (X^T X) (\hat{\beta} - \bar{\beta})$ to SSR.